

CAREERMATRI · CAREER DECISION BRIEF

Respondent

Mechanical Engineering · Bachelor's · BASIC · 2026-10-05

DECISION

The three to look at first

1

Mechanical Design & Product Engineering

Interest High

Experience High

Core 0/6 confirmed

Build evidence

2

CAE / Simulation / Computational Engineering

Interest High

Experience High

Core 0/9 confirmed

Build evidence

3

Thermal / Fluids / Energy Systems

Interest High

Experience High

Core 0/6 confirmed

Build evidence**What matters most in this result**

- The experience you recorded currently reaches Organisational value is confirmed.
- Mechanical Design & Product Engineering and CAE / Simulation / Computational Engineering sit close together, so this is the point to compare the two.

Your most important evidence gap**Interpreting requirements**

Mechanical Design & Product Engineering · Turned the given conditions into values you could work with, such as dimensions, loads and performance

What to do next

Interpreting requirements do one piece of work that fills that gap. Turned the given conditions into values you could work with, such as dimensions, loads and performance

Nothing here measures your ability or your odds of being hired. We wrote only what your answers and recorded experience confirm.

DECISION

Comparing the roles to look at first

Where the cells diverge is what tells you what to do next.

Role	Work style	Interest	Experience	Learning intent	Core confirmed	Evidence depth	Current
Mechanical Design & Product Engineering	Further away	High	High	High	0 / 6	E4 Organisational value	Build evid
CAE / Simulation / Computational Engineering	Further away	High	High	High	0 / 9	E4 Organisational value	Build evid
Thermal / Fluids / Energy Systems	Further away	High	High	High	0 / 6	—	Build evid
Worth keeping in view							
Automation / Robotics / Mechatronics	Further away	High	Low	Low	0 / 6	—	Look at th
Engineering Data / Digital Engineering	Further away	High	Medium	Low	0 / 6	E4 Organisational value	Look at th

We do not merge the cells into one. Where interest is high but experience is empty, doing one small thing first beats applying, and comes first. To say that, the two cells have to be separated. All sixteen roles are in the appendix.

WHY

how what you learned is used in real work

This runs in one line from what you learned to what the organisation counts as a result.

Mechanical Design & Product Engineering

Settling on one buildable form while the requirements pull against each other

What you learned

Mechanics of materials Statics

This is what you met in class. Having taken a course is not the same as being able to do it

Real work

turn the requirements into dimensions and tolerances

set the structure with the loads in mind

compare the design options and choose one

Technical judgement

Thickness and geometry

Material

Safety margin

Output

Design options and drawings that go on to production

3D models and drawings

Design calculation sheet

What the organisation values

Performance

cost

manufacturability

Private company is the basis.

My evidence today

bracket weight reduction capstone · Organisational value

Evidence still needed

A drawing or 3D model taken through to completion

A comparison table of design options

A record of building it and correcting it

EVIDENCE

The evidence I have today, and what is not yet confirmed

A lower rung does not mean something is lacking. It is the range confirmed by what you have recorded.

bracket weight reduction capstone

Activity	Confirmed	Turned the requirements into dimensions and compared three design options
Judgement	Confirmed	Picked the rib layout and thickness that cost less to machine · rib layout
Output	Confirmed	Drawing · analysis report
Performance	Confirmed	15% margin against the allowable stress in test · compared against the allowable stress
Organisational value	Confirmed	Adopted as the final design
Repeatability	Not yet	

Repeatability to get there Have you used the same method on another project

Not yet confirmed

This does not mean you cannot do it. It means that from what you have recorded, it is not confirmed.

Just pick from the options

What you pick here is not part of your fit. It only changes how far your evidence is confirmed.

Interpreting requirements · What judgement was that result used for

Whether the result led to a decision is what applications and interviews read.

What judgement was the result used for you may pick more than one

- ☐ Design option selection
 ☐ Dimensional change
 ☐ Material change
 ☐ Condition or setting change
 ☐ Deciding the test conditions
 ☐ Root cause analysis
 ☐ Decided whether to continue or stop
 ☐ It was not used directly in a decision

Who made that decision

- ☐ I decided
 ☐ I proposed it and we decided together
 ☐ Someone else decided
 ☐ It was already fixed

Generating and comparing design options · What did you weigh it against when choosing

It reads as your own decision only when there is a standard you compared against.

What did you compare the result against you may pick more than one

- ☐ Design criteria
 ☐ Experimental values
 ☐ An existing product
 ☐ An alternative design
 ☐ Theoretical values
 ☐ Another analysis method
 ☐ Codes and standards
 ☐ I did not compare

On what basis did you choose you may pick more than one

- ☐ Performance
 ☐ Weight
 ☐ Cost
 ☐ Difficulty of manufacture
 ☐ Schedule
 ☐ Safety margin
 ☐ Reliability
 ☐ It was fixed, so there was nothing to choose
 ☐ Something else

How far each role's core areas are confirmed

Role	Confirmed so far	Partially confirmed	Not yet confirmed	Fill this first
Mechanical Design & Product Engineering	0 / 6	0	6	Interpreting requirements · Generating and comparing design options

ACTION

The next experience to build

Do not start several things. Fill the one gap that is widest right now.

What is still empty

Interpreting requirements

Turned the given conditions into values you could work with, such as dimensions, loads and performance

- 1 something small you can scope: Mechanical Design & Product Engineering pick one problem on that side
- 2 write down the conditions and assumptions first
- 3 Interpreting requirements do the equivalent yourself
- 4 Performance · cost check it against
- 5 where it differed, write down why and correct it
- 6 leave it as a two-page record

What to leave behind Design options and drawings that go on to production · 3D models and drawings · why you chose it · what you compared it against · its limits

What to do, and when

30 days

- **Interpreting requirements** fill that one gap and nothing else. Turned the given conditions into values you could work with, such as dimensions, loads and performance
- take one small thing all the way through, inside a course or a club

This plan comes from the gaps you have right now. Record more experience and this line changes.

APPENDIX

These were left out of the main report. Use them when you want to trace something back.

All sixteen role families

The main report carries only the three to look at first and the two to keep in view.

Role	Work style	Interest	Experience	Learning intent	Core confirmed	Current stage
Mechanical Design & Product Engineering	Further away	High	High	High	0 / 6	Build evidence
CAE / Simulation / Computational Engineering	Further away	High	High	High	0 / 9	Build evidence
Thermal / Fluids / Energy Systems	Further away	High	High	High	0 / 6	Build evidence
Automation / Robotics / Mechatronics	Further away	High	Low	Low	0 / 6	Look at this first
Engineering Data / Digital Engineering	Further away	High	Medium	Low	0 / 6	Look at this first
R&D / Advanced Development	Further away	High	Low	Low	0 / 8	Look at this first
Systems Engineering / Technical PM	Further away	High	Low	—	0 / 6	Look at this first
Research Scientist / Academic Research	Further away	Medium	Medium	Low	0 / 7	Look at this first
Technical Consulting / Engineering Services	Further away	Medium	Low	—	0 / 6	Look at this first
Process / Industrialization Engineering	Further away	Medium	Low	Low	0 / 6	Look at this first
Materials / Tribology / Mechanical Failure	Further away	Low	Low	—	0 / 5	Set aside for now
Quality / Reliability / Failure Analysis	Further away	Low	Low	Low	0 / 6	Set aside for now
Equipment / Maintenance / Facilities	Further away	Low	Low	Low	0 / 5	Set aside for now
Test / Verification / Validation	Further away	Low	Low	Low	0 / 6	Set aside for now
Manufacturing / Production Engineering	Further away	Low	Low	Low	0 / 6	Set aside for now
Applications / Field / Customer Engineering	Further away	Low	Low	—	0 / 5	Set aside for now

which questions produced this line

The main report drops the question numbers and gives the reason in plain words. When you want to trace it back we leave them here for you.

Role	Questions behind it
Mechanical Design & Product Engineering	Q1, Q17, Q41, Q50, Q53
CAE / Simulation / Computational Engineering	Q2, Q18, Q42, Q50
Thermal / Fluids / Energy Systems	Q10
Automation / Robotics / Mechatronics	Q9, Q24, Q45
Engineering Data / Digital Engineering	Q15, Q22, Q46, Q50

Work style in detail

This axis rests on two questions, so we put no number on it. Until it is validated we describe it only as close, mixed or further away.

Axis	Leaning	Questions
Independent ↔ Collaborative	Neither side	2
Challenge ↔ Stability	Neither side	2
Speed ↔ Quality	Neither side	2

whether a result produced once can be reused

produced a result once

Confirmed

rocket weight reduction capstone

used the same method again

Not yet

carried it to another problem

Not yet

handed it over for others to use

Not yet

A degree does not raise it. It rises only from the projects and methods you recorded.

Full evidence coverage by role

Core, supporting and optional are all laid out. An empty optional area is never held against you.

Mechanical Design & Product Engineering

Core areas

Confirmed 0

Partial 0

Not yet 6

all 6

Confirmed so far

- ✓ Use of design tools Optional
bracket weight reduction capstone · Tools: SolidWorks

Not yet confirmed

- ☐ Interpreting requirements
- ☐ Generating and comparing design options
- ☐ Judging geometry, dimensions and material
- ☐ CAD and drawing outputs
- ☐ Structural and functional review
- ☐ Validation or design review
- ☐ Tolerances Supporting
- ☐ Manufacturability Supporting
- ☐ Cost Supporting
- ☐ Change control Supporting
- ☐ Interfaces Supporting
- ☐ Feedback from real build and test Supporting
- ☐ Industry standard Optional

CAE / Simulation / Computational Engineering

Core areas

Confirmed 0

Partial 0

Not yet 9

all 9

Confirmed so far

- ✓ Convergence check Supporting
bracket weight reduction capstone
- ✓ Use of analysis tools Optional
bracket weight reduction capstone · Tools: ANSYS Mechanical

Not yet confirmed

- ☐ Problem definition
- ☐ Model creation
- ☐ Assumptions set out
- ☐ Boundary conditions and loads
- ☐ Material and model parameters
- ☐ Running the analysis
- ☐ Interpreting the result
- ☐ Validation
- ☐ Link to a design decision
- ☐ Mesh independence Supporting
- ☐ Parametric study Supporting
- ☐ Automation and scripting Supporting
- ☐ Documenting the analysis Supporting
- ☐ Large-scale computation Optional
- ☐ Optimisation Optional

Thermal / Fluids / Energy Systems**Core areas**

Confirmed 0

Partial 0

Not yet 6

all 6

Not yet confirmed

- ☐ Load calculation
- ☐ System layout
- ☐ Selecting equipment and conditions
- ☐ Analysis and calculation results
- ☐ Performance check
- ☐ Reflected in the design
- ☐ Efficiency and energy Supporting
- ☐ Transient behaviour Supporting
- ☐ Operational control Supporting
- ☐ Test rig Supporting
- ☐ Renewables and power generation Optional
- ☐ HVAC design Optional

Automation / Robotics / Mechatronics**Core areas**

Confirmed 0

Partial 0

Not yet 6

all 6

Not yet confirmed

- ☐ Deciding the scope of automation
- ☐ Mechanism and cell design
- ☐ Actuator and sensor selection
- ☐ Control implementation
- ☐ Commissioning
- ☐ Automation result
- ☐ Safety design Supporting
- ☐ Vision and inspection Supporting
- ☐ Offline validation Supporting
- ☐ Return on investment Supporting
- ☐ A specific robot or PLC Optional
- ☐ Material handling automation Optional

Engineering Data / Digital Engineering**Core areas**

Confirmed 0

Partial 0

Not yet 6

all 6

Confirmed so far

- ✓ Programming language Optional
bracket weight reduction capstone

Not yet confirmed

- ☐ Choosing what to automate
- ☐ Data structure design
- ☐ Building tools and scripts
- ☐ Result validation
- ☐ How it was used
- ☐ Someone else used it
- ☐ Usage instructions Supporting
- ☐ Change history Supporting
- ☐ Connecting to an existing system Supporting
- ☐ Visualisation Supporting
- ☐ Predictive model Optional

We do not turn the number of confirmed cells into a single We do not merge them into a number. Merging hides what the other cells were. An empty optional area is never held against you.

how the material was produced

- Interest, experience, decision ownership, work style and learning intent come from different questions and are read separately. We never merge them into one score.
- Experience is not part of your fit. It only changes how far your evidence is confirmed, differ.
- The evidence ladder is how deep one experience goes; evidence coverage is how much of the role's territory it covers. We never merge the two.
- A degree does not raise your evidence rung.

Limits

- Cognitive interviews and the pilot are still ahead, so we do not report a measurement error.
- These are not measures of ability or of your odds of being hired.
- The analysis below is written against the role itself. We did not ask for a target country, so Country-specific material such as licensing, salary, hiring practice and work authorisation is not included. Written without verified data, someone will believe it and act on it.
- The sixteen role families and their evidence areas hold for mechanical engineering only.

Version

Assessment	ME_V2_DECISION_2026
Result schema	3.0
Evidence map	1.0
Questions	48 of 48 answers
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